3 (a) Complete this table of values for $y = x^2 + x - 4$

x	–3	–2	–1	0	1	2	3
y	2	–2	–4	–4	–2	2	8

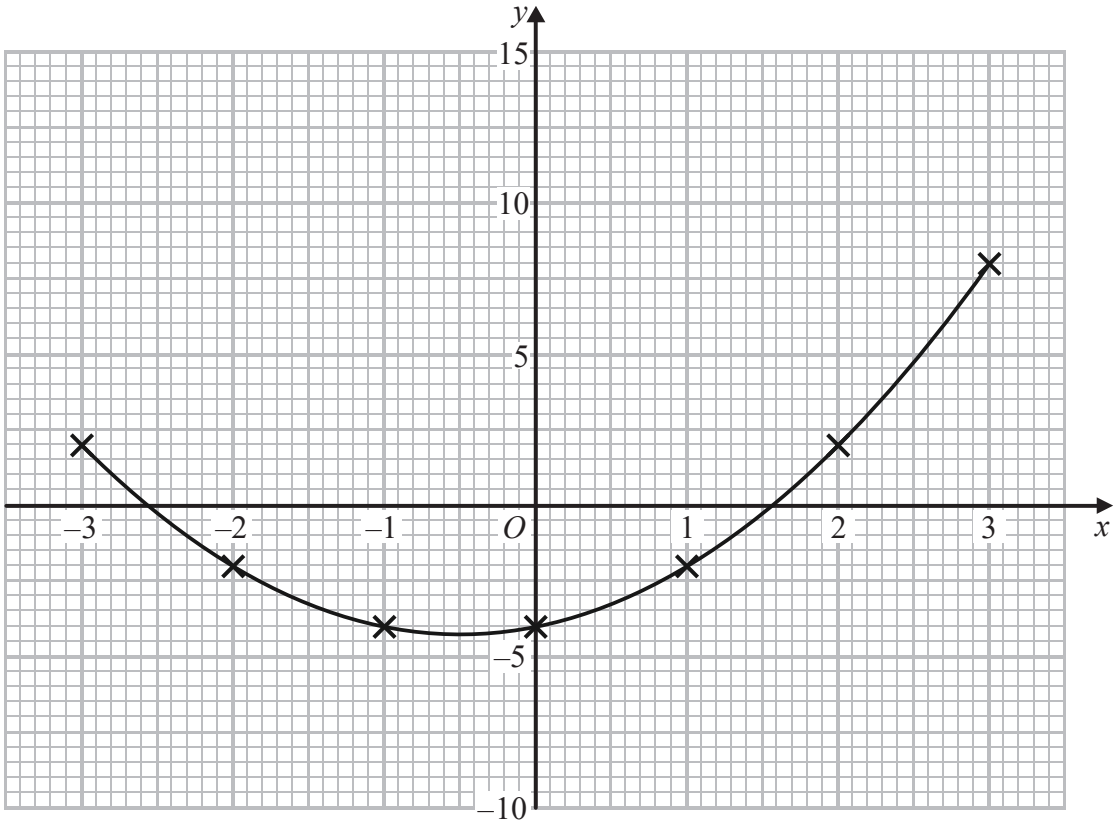
$(-3)^2 - 3 - 4 = 2$ $\quad$ $2^2 + 2 - 4 = 2$

$0^2 + 0 - 4 = -4$ $\quad$ $3^2 + 3 - 4 = 8$

2 marks total
All four missing values correct.

(2)

(b) On the grid, draw the graph of $y = x^2 + x - 4$ for values of x from –3 to 3



1 mark
At least five correctly plotted points.

1 mark
Smooth curve, including its minimum.

(2)

(c) Use the graph to estimate a solution to $x^2 + x - 4 = 0$

1 mark
Read a root where the curve crosses y = 0.

1.6 .....

(1)

(Total for Question 3 is 5 marks)

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